

TAKE HOME FINAL EXAMINATION

Course: **CHE 301. CHEMICAL ENGINEERING THERMODYNAMICS**
Term: Spring 2020
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Your name: _____
Due Date: Monday May 4th 2020 by 11:59 pm

Number of points.....100
Your score.....

INSTRUCTIONS FOR FINAL EXAMINATION DELIVERY

- Your Final Examination must be typed. No Exceptions.
- Scan your Final Examination and create a PDF file
- PDF File Name: FinalExamination_XY.PDF
 - **X:** First letter of your first name
 - **Y:** Your last name
 - Example: FinalExamination_MCaracotsios.PDF
- Upload your PDF file by the due date and time shown above

Note: All problems are worth 100 points. Your final score will be the average value

Problem 1: Tank Pressurization Calculations with the Redlich-Kwong EOS

A cylindrical tank with volume $\{V = 25\text{ m}^3\}$ is originally filled with nitrogen at conditions of $\{T_o = 25^\circ\text{C}$ and $P_o = 1\text{ bar}\}$. The tank must be pressurized to a pressure of $\{P = 25\text{ bar}\}$ using a high pressure line which supplies nitrogen at a constant temperature and pressure equal to $\{T = 25^\circ\text{C}$ and $P = 50\text{ bar}\}$.

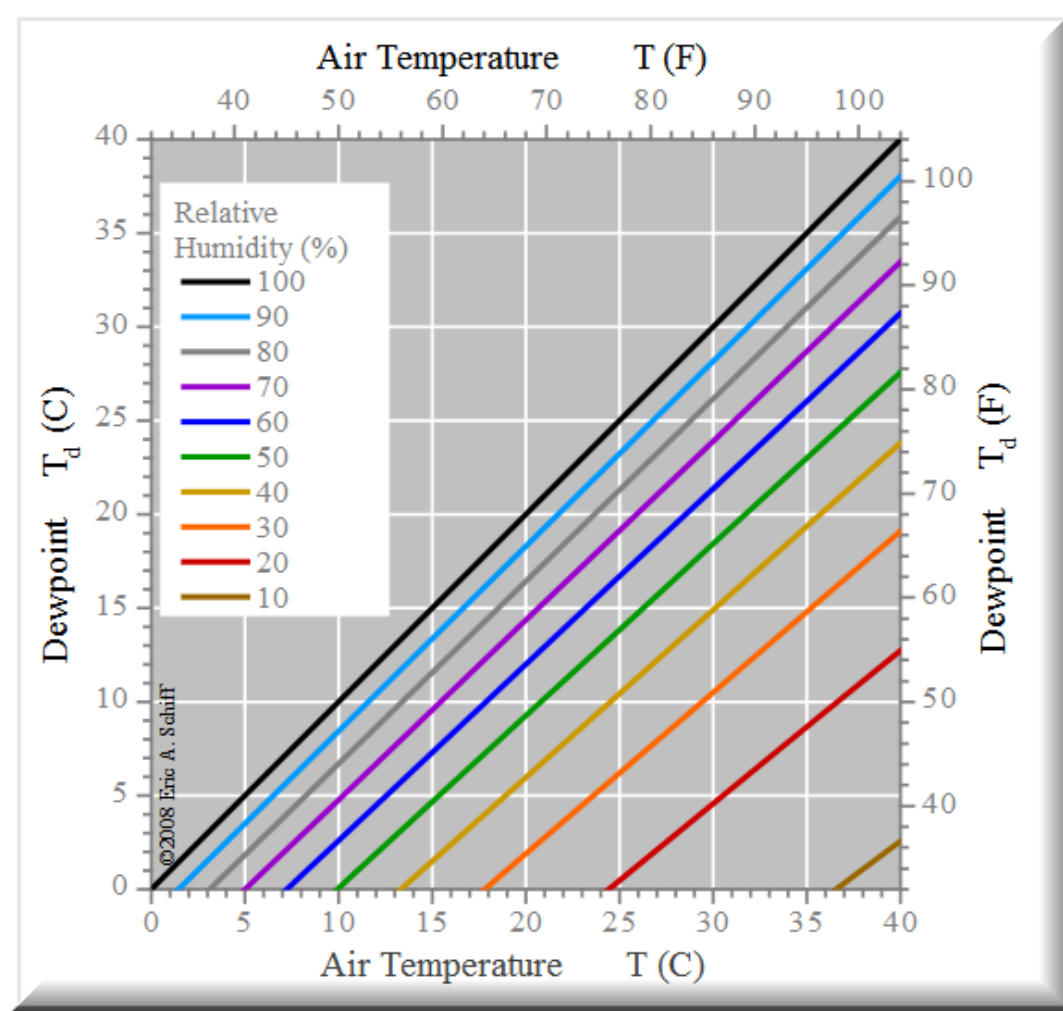
- Prove that by combining the dynamic energy and mass balances we obtain the equation:

$d(MU) = H dM$ where M is the mass(in moles) of nitrogen inside the tank at any time, U is the molar internal energy of nitrogen inside the tank and H is the molar enthalpy of the nitrogen in the high pressure line feeding the tank

- Integrate the equation obtained in Part 1. and calculate the amount of nitrogen added to the tank during the pressurization process using (a) the ideal gas law and (b) the Redlich-Kwong equation of state. What is the error in assuming that the gas is ideal.

Problem 2: Dew Point Calculations

On Tuesday July 8, 2003 in Dhahran Saudi Arabia, the recorded temperature at 12:00 noon was 104°F and the sea level pressure 29.39 inHg. The relative humidity was recorded at 72%. Estimate the dew point of the air from the chart given below, and verify your estimation using a thermodynamic model of your choice. If you decide to use Henry's law for oxygen, nitrogen, argon and carbon dioxide values for the Henry's law constants for these gases can be found in your notes. The composition of dry air can be found in the open literature.

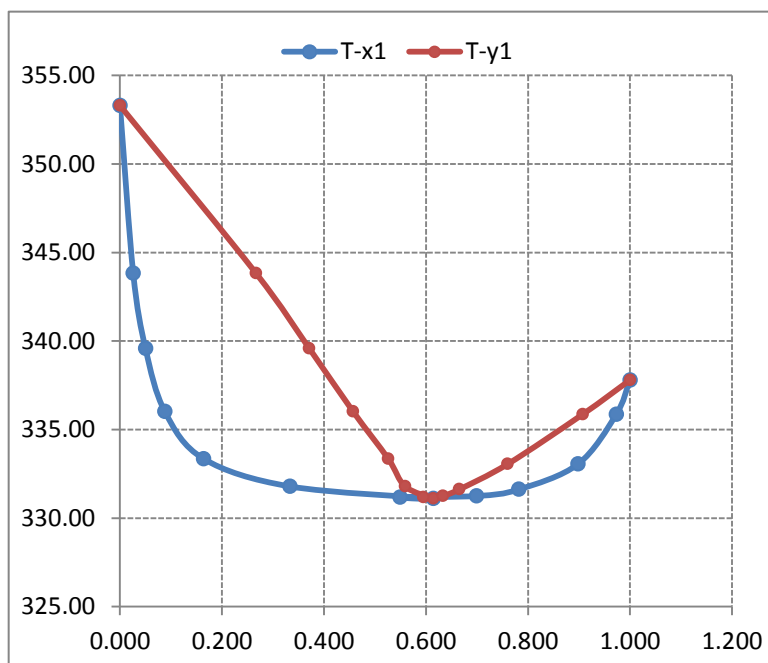


Problem 3: VLE Data Reduction with the Peng-Robinson Equation of State

Consider a mixture of Methanol (Component #1) and Benzene (Component #2) at 1.01325 bar that follows the Peng-Robinson Equation of State.

- Assume that the binary interaction parameter k_{12} is zero. Calculate the bubble point temperature at values of x_1 shown below and the dew point temperature at values of y_1 shown below. Plot the bubble point temperatures vs. liquid mole fraction x_1 and dew point temperature vs. vapor mole fraction y_1 . Superimpose the experimental data given below onto your graph and comment on the goodness of fit, by calculating the mean residual sum of squares. A value less than 10.0 indicates an acceptable result.
- At the azeotrope, the composition of liquid and vapor phases are the same at equilibrium. Use trial and error to determine the temperature and composition that is predicted for the azeotrope between methanol and benzene at 1.01325 bar if k_{12} is zero. Compare your result with the actual values given in the data below.
- Determine the optimal value of the binary interaction parameter k_{12} that can be used for the prediction of the azeotrope of this mixture

Temp(C)	x_1	x_2	y_1	y_2
80.15	0.000	1.000	0.000	1.000
70.67	0.026	0.974	0.267	0.733
66.44	0.050	0.950	0.371	0.629
62.87	0.088	0.912	0.457	0.543
60.20	0.164	0.836	0.526	0.474
58.64	0.333	0.667	0.559	0.441
57.95	0.614	0.386	0.614	0.386
58.02	0.549	0.451	0.595	0.405
58.10	0.699	0.301	0.633	0.367
58.47	0.782	0.218	0.665	0.335
59.90	0.898	0.102	0.760	0.240
62.71	0.973	0.027	0.907	0.093
64.65	1.000	0.000	1.000	0.000



Problem 4: Excess Enthalpy Model for Isopropanol-Water Mixtures

Your first project at the XYZ corporation is to develop a methodology for the calculation of the molar enthalpy of isopropyl alcohol(1)-water(2) liquid mixtures at different temperatures and pressure equal to 1 atm. There is a large number of the excess enthalpy experimental data available at different temperatures and mixture compositions and these are given in the table below: (all data are atmospheric pressure)

T=273.15K		T=308.15K		T=328.15K	
x_1	H^E , kJ/kmol	x_1	H^E , kJ/kmol	x_1	H^E , kJ/kmol
0.001	-20.934	0.0029	33.494	0.0047	-36.006
0.005	-97.134	0.013	-121.417	0.019	-115.137
0.014	-257.907	0.057	-355.878	0.063	-209.34
0.026	-439.614	0.103	-456.361	0.084	-234.461
0.083	-866.667	0.151	-439.614	0.1	-247.021
0.087	-870.854	0.351	-171.659	0.138	-198.873
0.135	-983.898	0.448	-33.494	0.158	-173.333
0.179	-921.096	0.577	157.424	0.163	-160.354
0.226	-862.481	0.673	211.015	0.25	-8.792
0.509	-401.933	0.767	217.714	0.359	146.538
0.704	-106.345	0.891	146.538	0.456	292.657
0.788	-9.211	0.929	103.833	0.615	410.306
0.864	33.494	0.969	48.148	0.637	418.68
0.904	30.982			0.789	319.871
0.939	22.609			0.903	185.057
				0.909	174.59
				0.967	69.92

- Develop a model for the Excess Enthalpy H^E using the thermodynamic principles you have learnt in class and estimate the parameters of the model from the experimental data.
- Using the model you developed for the excess enthalpy calculate the amount of heat in kJ per kmol of solution exchanged with the surroundings when 20 moles of isopropyl alcohol are mixed **isothermally** with 80 moles of water at 315.15 °K.
- If the pure components are initially at 315.15 °K and the mixing is **adiabatic** what is the final temperature of the solution.

Problem 5: VLE Data Reduction with the NRTL Model

The following P_{xy} data for a Toluene/n-Decane mixture are available at $T = 373.47^\circ\text{K}$:

TEMPERATURE	PRESSURE	X_1	X_2	Y_1	Y_2
K	Pa	TOLUENE	n-DECANE	TOLUENE	n-DECANE
373.47	9450	0.0000	1.0000	0.0000	1.0000
373.47	11630	0.0353	0.9647	0.2415	0.7585
373.47	14180	0.0700	0.9300	0.3834	0.6166
373.47	17240	0.1223	0.8777	0.5147	0.4853
373.47	22710	0.1880	0.8120	0.6544	0.3456
373.47	29530	0.3006	0.6994	0.7732	0.2268
373.47	34260	0.3665	0.6335	0.8277	0.1723
373.47	45240	0.5301	0.4699	0.9029	0.0971
373.47	50040	0.6144	0.3856	0.9251	0.0749
373.47	58680	0.7769	0.2231	0.9560	0.0440
373.47	66370	0.8617	0.1383	0.9746	0.0254
373.47	75380	1.0000	0.0000	1.0000	0.0000

From the **Gibbs Excess** energy function for the **NRTL (Non-Random Two Liquid)** model the liquid activity coefficients for the two mixture components are given by:

$$\ln \gamma_1 = x_2^2 \left[\tau_{21} \left(\frac{G_{21}}{x_1 + x_2 G_{21}} \right)^2 + \frac{\tau_{12} G_{12}}{(x_2 + x_1 G_{12})^2} \right]$$

$$\ln \gamma_2 = x_1^2 \left[\tau_{12} \left(\frac{G_{12}}{x_2 + x_1 G_{12}} \right)^2 + \frac{\tau_{21} G_{21}}{(x_1 + x_2 G_{21})^2} \right]$$

where

$$\ln G_{12} = -0.3\tau_{12} \text{ and } \ln G_{21} = -0.3\tau_{21}$$

■ Prove the following functional expressions:

$$\ln \gamma_1^\infty = \tau_{21} + \tau_{12} \exp(-0.3\tau_{12})$$

$$\ln \gamma_2^\infty = \tau_{12} + \tau_{21} \exp(-0.3\tau_{21})$$

■ Determine whether that experimental data are thermodynamically consistent.

■ Estimate from the experimental data the **NRTL** model coefficients τ_{12} , G_{12} and τ_{21} , G_{21} and draw a parity plot of predicted pressure values vs. experimental values. (Hint: Use the expressions for liquid activity coefficients at infinite dilution to obtain initial guesses for the constants)